Minh-Ha Nguyen

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Trained in pharmacology and clinical pharmacy. Building agentic systems with statistical and clinical judgement.

Employment

Graduate Researcher (PhD) Vanderbilt University Medical Center August 2024–present

Designed and built **PIHF-MCP**, a self-improving multi-agent harness translating scientific intent into executable workflows, with persistent state, error recovery and expert feedback (*arXiv preprint, 2026*).

Developed **LiteOdyssey**, translating clinical questions into phenotype-to-gene prioritization, disease-evidence review and literature retrieval, improving rare-disease diagnosis on held-out cases across benchmarks and models (*arXiv preprint, 2026*).

Investigated conflicting findings in the medical literature and offered unified explanations and solutions using experimental design methods: large language model performance in rare diseases (*npj Digital Medicine, 2026*) and risk of depression with 5-alpha-reductase inhibitors (*Postgraduate Medicine, 2026*).

Visiting Research Fellow Yonsei University College of Medicine January–September 2023

Big Data Epidemiology

Designed nationwide new-user studies of prostate drugs and diabetes outcomes, using causal inference to make treatment groups more comparable.

Published in *Yonsei Medical Journal, 2026* and *Heliyon, 2025*.

Junior Lecturer and Clinical Researcher January 2020–June 2024

University of Medicine and Pharmacy at Ho Chi Minh City · School of Pharmacy

Taught pharmacy practice and interprofessional education.

Carried out research on medical education and antibiotic stewardship using machine learning; first-prize oral presentation on low-cost antibiotic-stewardship ML at AASP 10 (2022).

Education

PhD Candidate, Epidemiology Vanderbilt University 2024–2027 (expected)
Vanderbilt University School of Medicine

NASA-supported doctoral research: low-dose radiation and Parkinson's disease.

MPharm, Pharmacy Queen's University Belfast 2018

Awards and Grants

Anthropic AI for Science grant: DisMechOdyssey research team	2026
Vanderbilt Graduate Fellowship	2024
MEF Fellowship by KOICA	2023
AASP 10: First-prize oral presentation	2022

Selected Publications

From Answers to Policies: Efficient In-Context Learning System through Emulating Expert Investigation.
arXiv preprint, 2026.

Teaching agentic AI to generalize expert diagnostic reasoning in rare diseases.
arXiv preprint, 2026.

Diagnostic accuracy of large language models for rare diseases: a systematic review and meta-analysis.
npj Digital Medicine, 2026.

Professional Service and Skills

GBD Senior Collaborator: Global Burden of Disease Collaborator Network (IHME)

Peer reviewer (2025–2026): *Diabetes, Obesity and Metabolism*; *Therapeutic Advances in Drug Safety*
Agentic Engineering: orchestration, MCP/API integration, evaluation pipelines, local inference (SGLang)

Scientific Computing: Python, R, SAS

Statistics & Epidemiology: statistical inference, causal inference, experimental design